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Detecting the Cure Model Appropriateness in Randomized Clinical Trials With Long-Term Survivors

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ABSTRACT

PURPOSE To evaluate the appropriateness of a cure model when analyzing right-censored end points of a randomized clinical trial (RCT) in malignancy in the presence of long-term survivors. We aim to derive how the ratio estimation of censored cured subjects (RECeUS), previously proposed for a homogeneous population, could be extended for use in RCTs.

METHODS Based on the RECeUS method, four decision rules were considered to assess the appropriateness of a cure model. They considered the eligibility conditions to be met: in both arms, in at least one randomized arm, in the entire sample, or when only considering an average of the conditions, respectively. A simulation study was performed to evaluate their performance and the impact of the link function when considering the appropriateness of cure models. We also illustrate the method using two real data examples from two RCTs conducted in patients with acute leukemia and COVID-19 disease.

RESULTS Simulation results show that the best decision rule that can be applied in all considered treatment effect scenarios might be to check the criteria in at least one randomized arm. Regardless of the rules, the cure model appeared to be appropriate in both RCT data.

CONCLUSION When analyzing survival data from RCTs, the appropriateness of a cure model could be considered in the face of a plateau shape of the survival curves. To ensure that the presence of such a plateau in the survival curves is a reliable indicator of the presence of cured patients in the population, the RECeUS method should be used in each randomized arm separately, with criteria met in at least one randomized arm.

ACCOMPANYING CONTENT

Data Supplement

Protocol

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INTRODUCTION

In randomized clinical trials (RCTs) conducted in patients with hematologic malignancies, survival is often the primary end point used to assess the effect of treatment. Other event times are also used as primary or secondary end points, such as relapse-free survival (RFS). The typical analysis of these data is based on nonparametric estimation of survival in the randomized groups, compared using the log-rank test, with estimation of the treatment effect size from a Cox model.

These standard survival analyses assume that all patients in the trial will eventually experience the event of interest. However, if the treatment effect is significant, the risk of relapse or death for some patients may become very small.[1,2] This may indicate heterogeneity within a patient population, with a proportion of long-term survivors (for whom the risk of relapse or death become negligible after a certain point) potentially illustrated by the plateau shape of the survival curve and a proportion of patients who will succumb to the disease. In this setting, cure models have been proposed to investigate which covariates are associated with short-term or long-term effects.[3] Two main types, mixture cure models[4,5] and nonmixture cure models, have been proposed.[4,6,7] Their application to patient-level RFS data from two RCTs recently demonstrated that the RFS gains with nivolumab and ipilimumab in patients with resected melanomas were primarily due to more patients being long-term recurrence-free survivors.[8] These types of findings can provide information that can be useful to help patients understand the trade-offs with different therapies.

However, the presence of such a plateau in the survival curves is not always a reliable indicator of the presence of cured patients in the population. Such plateaus can be induced by administrative censoring; indeed, large errors in estimates can result from limited clinical data, either small samples or short follow-up. The instability of small event numbers in analyses with incomplete follow-up and the possibility of many additional events with additional follow-

up is one potential explanation for the lack of survival benefit observed in confirmatory phase III trials.[9] The administrative censoring may also bias the estimate of the proportion cured by the cure models. Insufficient follow-up could lead to inaccurate estimates of the cure model, as recently reported in a *Journal of Clinical Oncology* editorial.[10] This has led to the development of several methods although they are subject to limited statistical power and increased type I error.[11-13] Even in the presence of cure proportions, these approaches fail to identify that a cure fraction that exists. To address these limitations, a new method, the ratio estimation method of censored uncured subjects (RECeUS), is proposed to estimate the proportion of noncured patients at the end of the study to measure the sufficiency of the follow-up.[14] In simulations, the method was able to successfully identify scenarios in which the cure model was inappropriate because of short follow-up or high administrative censoring.

However, all existing methods have considered a homogeneous population. The aim of this article was to evaluate the relevance of the RECeUS in the analysis of RCTs. A simulation study is provided, evaluating when the population is heterogeneous in terms of either the proportion of cured patients or the survival of sensitive patients. We used two real data sets from two RCTs conducted in patients with ALL or COVID-19 disease as illustrative examples.

METHODS

Evaluating Mixture Cure Model Appropriateness in Randomized Clinical Trials

Mixture Cure Models

We considered a n-sample of independent observations (t_i, δ_i, x_i), where t_i is the time to the event of interest observed for individual i, $\delta_i \in 0,1$ is the censoring indicator (1 if the observation is not censored, 0 otherwise), and x_i is the vector of baseline characteristics of patients at inclusion in the study. We define U, the latent variable indicating the patient susceptibility to the event of interest: $U = 1$ when the patient could present the event, $U = 0$ otherwise (ie, cured). To analyze RCTs with cured patients, the overall survival function in a mixture cure model is defined as follows:

$$S(t|x) = \pi(x) + (1-\pi(x))S(t|U=1;\, x)$$

where $S(t|x)$ is the survival function of the event time T for the population, $S(t|U = 1;\, x)$ is the survival function of the patients at risk of the event (often called the latency of the model), and $\pi(x) = P(U = 0|x)$ is the probability of patients being cured conditional on the observed covariate vector x (often called the incidence of the model). In the context of a RCT, the covariate of interest is the random assignment group (let the arms be labeled A or B). In addition, the survival function of cured patients is set to 1 since, as postulated by the cure model, these patients will not experience an event.

For the population survival curve, this means that $S(t|x) \underset{t \to +\infty}{\to} \pi(x)$. The instantaneous hazard function becomes

$$h(t|x) = \frac{(1-\pi(x)) \times f(t|U=1;x)}{S(t|x)},$$

where $f(t|U = 1;\, x)$ is the conditional density function of T for noncured patients.

The estimation of $S(t|U = 1;\, x)$ can be performed in nonparametric or parametric frameworks, where the Weibull distribution is commonly used.[3,5,15,16] It is then possible to make its parameters dependent on covariates, for example, using a proportional hazards model.

Two main link functions are used to model the proportion cured, $\pi(x)$, the logistic function or the log-log link function, although others could be used. Previous authors have described patterns of sensitivity to the survival distribution,[17] and the 2-arm RCT setting ensures a simple form of covariate specification. We assessed the impact of the link function when considering the appropriateness of cure models.

Evaluating the Appropriateness of a Cure Model

Cure models are considered only valid when two fundamental conditions are met.[11]

Condition of a non-null cure fraction—To test the existence of cured patients in the population, a test statistic d_n was developed, based on a deviance statistic, distributed under H_0 according to a 50-50 mixture of a χ^2 distribution with 1 degree of freedom and a mass of 1.[18]
Condition of sufficient follow-up—A sufficient follow-up is the one for which we could observe the event time of all uncured patients, meaning that the right end of the censoring time distribution is greater than the right end of the survival time distribution for uncured or susceptible patients.

Tests to verify this condition are rare in the statistical literature. One nonparametric test was proposed,[11] which leads to a high risk of Type I error. Several modifications have been proposed based on the difference in follow-up between uncured and censored patients,[11,18,19] although unsatisfactory because of the instability of the Kaplan-Meier estimator in the presence of censoring. In practice, many studies do not have long enough follow-up to observe the full distribution of events in uncured patients (who are therefore censored). We proposed to use a recently developed method.[14]

Evaluating the Two Conditions Simultaneously: The RECeUS Method

The RECeUS method is based on an estimate of the proportion r of patients not cured at the time of analysis τ to

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quantify sufficient follow-up and on an estimate of the proportion cured to test for the presence of cure

$$\hat{r} = \frac{\hat{S}(\tau|U=1)}{\hat{S}(\tau)} = \frac{\hat{S}(\tau|U=1)}{\hat{\pi}_n + \left(1-\hat{\pi}_n\right)\hat{S}(\tau|U=1)}.$$

Heuristically, this quantifies the estimated proportion of noncured individuals at time τ, normalized to the estimated total proportion of patients censored at that time.

Two steps are required to test the relevance of a cure model using the RECeUS method after estimating $S(\tau)$, $S(\tau|U=1)$, and $\hat{\pi}_n$ from one parametric model on the sample of size n. One could also fit multiple models and use the AIC best fitting model.[14]

1. If $\hat{\pi}_n$ is sufficiently small ($<\alpha_\pi$), the cure model is probably not an appropriate model or the follow-up is not sufficient to obtain valid results.
2. If $\hat{\pi}_n \geq \alpha_\pi$, a small value of $\hat{r}_n$ ($\leq \alpha_{r_n}$) may indicate sufficient follow-up.

Decision thresholds, α_π and α_{r_n}, for the adequacy of a cure model were set to 2.5% and 5%, respectively.[14] When both conditions are met, a cure model is appropriate to analyze the survival of the studied population from data collected at analysis time τ on a sample of size n.

In the context of a randomized therapeutic trial, our question was whether these criteria should apply together or separately across arms. Thus, we derived four decision rules to assess the relevance of the cure model for the RECeUS method:

1. Both arms: If both eligibility conditions are met in both random assignment arms:
   $\hat{\pi}(x|A) \geq \alpha_\pi, \hat{r}_{n_A}|A \leq \alpha_{r_n}$ AND $\hat{\pi}(x|B) \geq \alpha_\pi, \hat{r}_{n_B}|B \leq \alpha_{r_n}$.
2. At least one arm: If the eligibility conditions are met in at least one arm.
   $\hat{\pi}(x|A) \geq \alpha_\pi, \hat{r}_{n_A}|A \leq \alpha_{r_n}$ OR $\hat{\pi}(x|B) \geq \alpha_\pi, \hat{r}_{n_B}|B \leq \alpha_{r_n}$.
3. On average: If the average of the statistics meets the predefined thresholds
   $\bar{\pi} \geq \alpha_n, \bar{r}_n \leq \alpha_{r_n}$ with $\bar{\pi} = \frac{\hat{\pi}(x|A) + \hat{\pi}(x|B)}{2}$ and $\bar{r}_n = \frac{\hat{r}_{n_A}|A \leq \alpha_{r_n} + \hat{r}_{n_B}|B \leq \alpha_{r_n}}{2}$.
4. Overall: When the conditions are met globally in the population, that is, without taking the treatment group into account:
   $\bar{\pi} \geq \alpha_n, \bar{r}_n \leq \alpha_{r_n}$, with $n = n_A + n_B$.

Illustrative Examples From to Real Clinical Settings

We illustrated the method using data from two multicenter RCTs. The GRAAPH-2014 trial included a total of 155 patients with Philadelphia-positive (ClinicalTrials.gov identifier: NCT02611492) ALL, randomly assigned between standard intensive chemotherapy (n = 76, arm A) and reduced intensity chemotherapy (n = 79, arm B).[20] We focused on RFS as the end point of interest. The 4-year estimate of RFS was 76.2% (95% CI, 67.1 to 86.4) in arm A and 58.5% (95% CI, 48.4 to 70.7) in arm B (hazard ratio [HR], 1.81 [95% CI, 1.03 to 3.20]; $P = .029$; Fig 1A). A plateau shape was observed in the 2 RFS curves after follow-ups of approximately 40 months in arm A and 60 months in arm B. This raises the question of whether these shapes could indicate the existence of cured and noncured patients or are simply the result of censoring related to the end of the study.

The COVIDICUS trial was conducted in patients with severe acute hypoxemic respiratory failure caused by COVID-19 pneumonia requiring intensive care unit (ClinicalTrials.gov identifier: NCT04344730) admission. A total of 546 patients were randomly assigned to standard (n = 276) or high-dose (n = 270) dexamethasone.[21] The outcome was time to all-cause mortality, assessed at day 60 after random assignment, with no difference between arms (HR, 0.96 [95% CI, 0.69 to 1.33]; $P = .79$). A plateau shape was observed in both randomized arms, suggesting the existence of cured (discharged alive) and not cured (who will die) patients.

We applied the RECeUS method and other existing test statistics[11] to these two data examples (Table 1). Depending on the method, the decision regarding the appropriateness of the cure model was possibly only suggested in one arm (GRAAPH-2014) or overall (COVICUS). To gain further insight into the criteria for evaluating whether the RECeUS method correctly concludes that a mixture cure model is appropriate in a RCT setting, we performed a simulation study.

Simulation Study

We considered a 1:1 randomized controlled trial with two parallel arms, A and B. The sample size was set to n = 280 to have an 80% power to detect an overall treatment effect on survival, resulting from the data generated in the first scenario. The study will run for 84 months, with uniform enrollment of patients over the first 40 months. Second, we used other patterns of follow-up, over 24 or 96 months.

Several effects were considered. First, there was no effect of treatment on either cure rate or survival, defining the benchmark. We then added a treatment effect, mimicking data from the previously described clinical trial in ALL,[20] in which treatment had no effect other than on the proportion cured (scenario 1). We then reversed the parameters to explore three other realistic scenarios (Data Supplement, Table S1). Sensitivity to sample size was assessed by varying n from 140 to 560.

Event times were generated using a mixture cure model, where the survival time of susceptible patients follows a Weibull PH distribution with the shape parameter set to 1.18 and scale parameter dependent on the treatment arm using a log link function and the cure rate was modeled with a treatment-dependent logistic link function. We examined deviations from the model to explore the impact of the specification of the cure fraction link function on the performance of the RECeUS method, by generating log-log and

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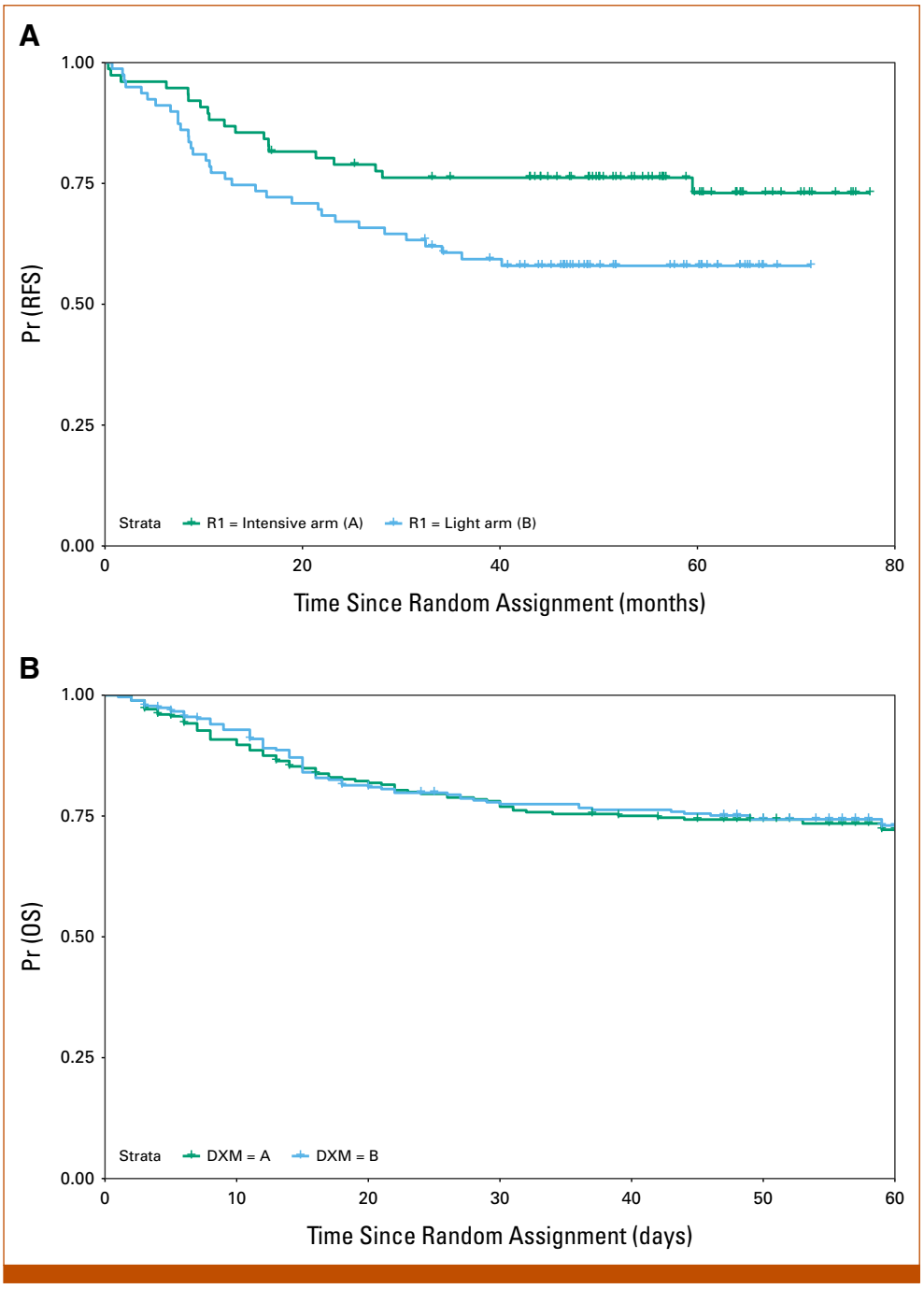

FIG 1. Illustrative examples: Main outcome of the GRAAPH-2014 (RFS, left plot) and the COVIDICUS (in-hospital death, right plot) trials. OS, overall-survival; RFS, relapse-free survival.

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probit links. We also used distinct Weibull shape parameters in the treatment arm to explore the influence of non-proportional hazards.

In each n-sample, we applied the four RECeUS decision rules described above to conclude whether the cure model was appropriate, with α_{π} and α_{r_n} set to 2.5% and 5%, as previously recommended.[14] For each scenario and each decision rule, we computed the proportion of incorrect conclusions of the cure model over the N = 10,000 simulations with 95% CIs.

RESULTS

Capacity to Detect the Appropriateness of the Cure Model

In the benchmark, the RECeUS method performed well in concluding that the cure model was appropriate, except for the choice of log-log modeling for the cure rate (Table 2). The error rates obtained varied depending on the decision rule. As expected, given the homogeneous population, the "at least one arm" or "overall" decision rules resulted in the lowest error rates (Table 2).

TABLE 1. Application of the RECeUS Approach Overall and Stratified by Arm of the Two Randomized Clinical Trial Data Illustrative Examples

	Uncured		Cured		Test Statistics	
Trial	$\hat{r}_n$ (95% CI), %	HR (95% CI)	$\hat{\pi}_n$	OR (95% CI)	$\hat{\alpha}_n$	$\hat{d}_n$
GRAAPH-2014 (by arm)						
Arm *A*	0.47	1.00	0.74 (0.63 to 0.83)	1.00	0.37	605.17
Arm *B*	0.75	1.06 (0.53 to 2.12)	0.57 (0.45 to 0.65)	0.46 (0.23 to 0.94)	<0.01	
Overall	0.43		0.65 (0.57 to 0.73)		0.37	612.22
COVIDICUS (by arm)						
Arm *A*	3.49	1.00	0.71 (0.66 to 0.77)	1.00	0.05[a]	2,090.81
Arm *B*	3.89	0.97 (0.64 to 1.47)	0.72 (0.66 to 0.78)	1.04 (0.86 to 1.54)	0.05[a]	
Overall	3.67		0.72 (0.68 to 0.76)		<0.01	2,094.46

NOTE. $\hat{r}_n$ and $\hat{\pi}_n$ refer to the RECeUS estimates of the proportion of uncured and cured patients, respectively, at the administrative censoring time; $\hat{\alpha}_n$ and d_n are the test statistics for sufficient follow-up and cured conditions, respectively, from the study by Maller and Zhou[18] (when $\hat{\alpha}_n \geq 0.05$, the hypothesis of sufficient follow-up is rejected; when $d_n \geq 2.71$, the hypothesis of no proportion of cured patients in the population is rejected).
Abbreviation: RECeUS, ratio estimation of censored cured subjects.
[a]Values very close but slightly below 0.05.

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In Scenario 1, where treatment has a significant effect on cure rate only, the best decision rules for determining cure model appropriateness were the "at least one arm" and "overall" rules. As above, the choice of a log-log link function to model the cure rate leads to a false conclusion of inappropriateness of the cure model regardless of the decision rule, except for the overall decision rule where the error rate was rather low at 0.89% (95% CI, 0.70 to 1.08). The error rates ranged from 0.22% (95% CI, 0.13 to 0.31) to 9. 36% (95% CI, 8.79 to 9.93) for the logit link function and from 0.64% (95% CI, 0.49 to 0.80) to 11.54% (95% CI, 10.91 to 12.17) for the probit link function, depending on the decision rules. Similar results were found when the cure rate increased from 0.25 to 0.50 (scenario 5), where the choice of the log-log function strongly affected the false discovery of an inappropriate cure model in the overall rule.

In Scenario 2, when treatment had a significant effect only on the survival of those not cured ($HR_u = 0.54$), the error rates again varied according to the link function used to model the cure rate and the decision rules. The log-log link function resulted in particularly high error rates, ranging from 19.64% (95% CI, 18.83 to 20.44) for the decision rule based on "at least one arm" to 76.96% (95% CI, 76.10 to 77.81) for the decision based on "both arms," with a model convergence rate of 93.14%. However, the "at least one arm" decision rule achieved moderate error rates, ranging from 3.67% (95% CI, 3.30 to 4.04) to 19.64% (95% CI, 18.83 to 20.44).

When the treatment has a very small effect on either the proportion cured or the survival time of those not cured (scenario 3), the results were, unsurprisingly, very similar to those observed in the benchmark, with relatively low error rates compared with the other scenarios, regardless of the link function or decision rule.

Finally, when the treatment affects both components (scenarios 4 and 6), the error rates obtained were particularly high for all decision rules and link functions used to model the cure rate.

However, by selecting the "at least one arm" decision rule, the error rates decreased and, as expected, were improved with a logistic link function.

When all scenarios were considered, the "at least one arm" or "overall" decision rules allowed the lowest error rates in concluding that the cure model was inappropriate for all link functions except the log-log, which had high error rates regardless of the scenario. The situations with the lowest error rates were those in which the cure rate was highest (at 0.75), that is, the benchmark and Scenario 3. These conclusions were robust to the use of data generated by a probit link. The results were similar when cure rates decreased, either because of a negative treatment effect in a population with a high baseline cure rate or because of a positive treatment effect in a population with a low baseline cure rate. By contrast, the performance of the "at least one arm" and "overall" decision rules decreased when the survival of the noncure population was prolonged.

Finally, the decision to reject or not reject a cure model was made on the basis of both criteria ($\hat{r}_n < 0.05$ and $\hat{\pi} > 0.025$). Looking at these two criteria separately, the decision to reject a model was almost always based on insufficient follow-up ($\hat{r}_n < 0.05$; Data Supplement, Table S2).

Sensitivity Analyses

Choice of the Log-Log Link Function

Given that the performance of the method was affected by the misspecification of a log-log link on the cure rate, we

TABLE 2. Simulation Study: Error Rates of Rejecting the Cure Model by Scenario and the Decision Rule

Decision Rules					
	Both Arms	On Average	At Least One Arm	Overall	
Scenarios	Proportion of Cure Model Rejections (95% CI), %				Convergence Rate
Benchmark					
Logit	3.13 (2.79 to 3.47)	1.96 (1.69 to 2.23)	0.39 (0.27 to 0.51)	0.25 (0.15 to 0.35)	100
Log-log	21.79 (20.97 to 22.60)	15.87 (15.14 to 16.59)	7.30 (6.79 to 7.81)	5.13 (4.70 to 5.57)	98.77
Probit	10.73 (10.12 to 11.34)	9.26 (8.69 to 9.83)	2.69 (2.38 to 3.01)	2.38 (2.08 to 2.68)	99.9
Scenario 1					
Logit	9.36 (8.79 to 9.93)	8.83 (8.28 to 9.39)	0.87 (0.69 to 1.05)	0.22 (0.13 to 0.31)	99.97
Log-log	29.50 (28.58 to 30.42)	25.74 (24.86 to 26.62)	21.62 (20.79 to 22.45)	0.89 (0.70 to 1.08)	94.68
Probit	11.54 (10.91 to 12.17)	10.80 (10.19 to 11.41)	4.57 (4.16 to 4.99)	0.64 (0.49 to 0.80)	99.46
Scenario 2					
Logit	40.41 (39.45 to 41.37)	23.42 (22.59 to 24.25)	3.67 (3.30 to 4.04)	10.39 (9.79 to 10.99)	99.98
Log-log	76.96 (76.10 to 77.81)	63.39 (62.41 to 64.37)	19.64 (18.83 to 20.44)	41.16 (40.16 to 42.16)	93.14
Probit	48.01 (47.03 to 48.99)	32.57 (31.65 to 33.49)	12.19 (11.55 to 12.84)	13.70 (13.02 to 14.37)	99.65
Scenario 3					
Logit	2.31 (2.02 to 2.60)	1.47 (1.23 to 1.71)	0.27 (0.17 to 0.37)	0.22 (0.13 to 0.31)	100
Log-log	19.04 (18.27 to 19.82)	14.09 (13.40 to 14.77)	6.47 (5.99 to 6.96)	4.19 (3.80 to 4.58)	99.04
Probit	9.11 (8.55 to 9.67)	7.94 (7.41 to 8.47)	1.60 (1.36 to 1.85)	2.97 (2.64 to 3.31)	99.89
Scenario 4					
Logit	53.28 (52.31 to 54.26)	32.80 (31.88 to 33.72)	8.43 (7.89 to 8.97)	17.63 (16.88 to 18.38)	99.88
Log-log	77.70 (76.85 to 78.54)	61.90 (60.91 to 62.89)	21.08 (20.25 to 21.91)	46.13 (45.11 to 47.14)	92.81
Probit	66.32 (65.39 to 67.25)	51.20 (50.22 to 52.19)	29.27 (28.37 to 30.16)	29.19 (28.30 to 30.09)	99.23
Scenario 5					
Logit	8.60 (8.05 to 9.15)	6.76 (6.27 to 7.25)	0.27 (0.17 to 0.37)	0.24 (0.14 to 0.34)	99.98
Log-log	8.03 (7.50 to 8.57)	7.36 (6.84 to 7.87)	3.52 (3.15 to 3.88)	77.27 (76.44 to 78.09)	99.24
Probit	12.01 (11.38 to 12.65)	11.37 (10.75 to 11.99)	1.61 (1.36 to 1.85)	0.65 (0.49 to 0.81)	99.55
Scenario 6					
Logit	63.95 (63.01 to 64.89)	43.83 (42.86 to 44.80)	15.53 (14.82 to 16.25)	12.86 (12.20 to 13.52)	99.84
Log-log	68.02 (67.10 to 68.93)	37.73 (36.78 to 38.69)	1.69 (1.44 to 1.95)	28.46 (27.57 to 29.35)	99.3
Probit	73.23 (72.36 to 74.10)	56.98 (56.00 to 57.95)	35.56 (34.62 to 36.50)	12.24 (11.60 to 12.89)	99.32

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wondered whether the correct specification of the link function might improve the results. However, the method still performed poorly, with an error rate of 20.22% (95% CI, 19.41 to 21.03) under the "at least one arm" decision rule, whereas the results obtained using the mis-specified log-log link with data generated using logit and probit link functions show low error rates (1.00%, 95% CI, 0.81 to 1.20 and 4.32%, 95% CI, 3.92 to 4.72). This means that even when the data are generated using a log-log link, the use of a model with logit or probit links outperforms the correctly specified log-log link model.

Effect Size on the Cure Fraction

Regardless of the effect of treatment on the cure fraction, the error rate remains below 10% for all decision rules, with the "overall" and "at least one arm" decision rules with error rates close to zero (Fig 2). Model performance was affected by the misuse of a log-log function, as previously.

Effect Size on the Survival of Susceptible

When the treatment has a deleterious effect on the noncured, thereby increasing the risk of an event in the latter subset, the RECeUS method increased the probability of correctly detecting the appropriateness of the cure model (Fig 3). However, when the treatment has a beneficial effect on the noncures, that is, prolongs survival of the noncure subset (the HR tends to 0.5), the method has difficulty in detecting the appropriateness of using a cure model.

The best performing decision rule was the "at least one arm" rule, followed by the "overall" decision rule, which performed similarly. The worst performing decision rule was "both arms," with error rates approaching 50% when the HR

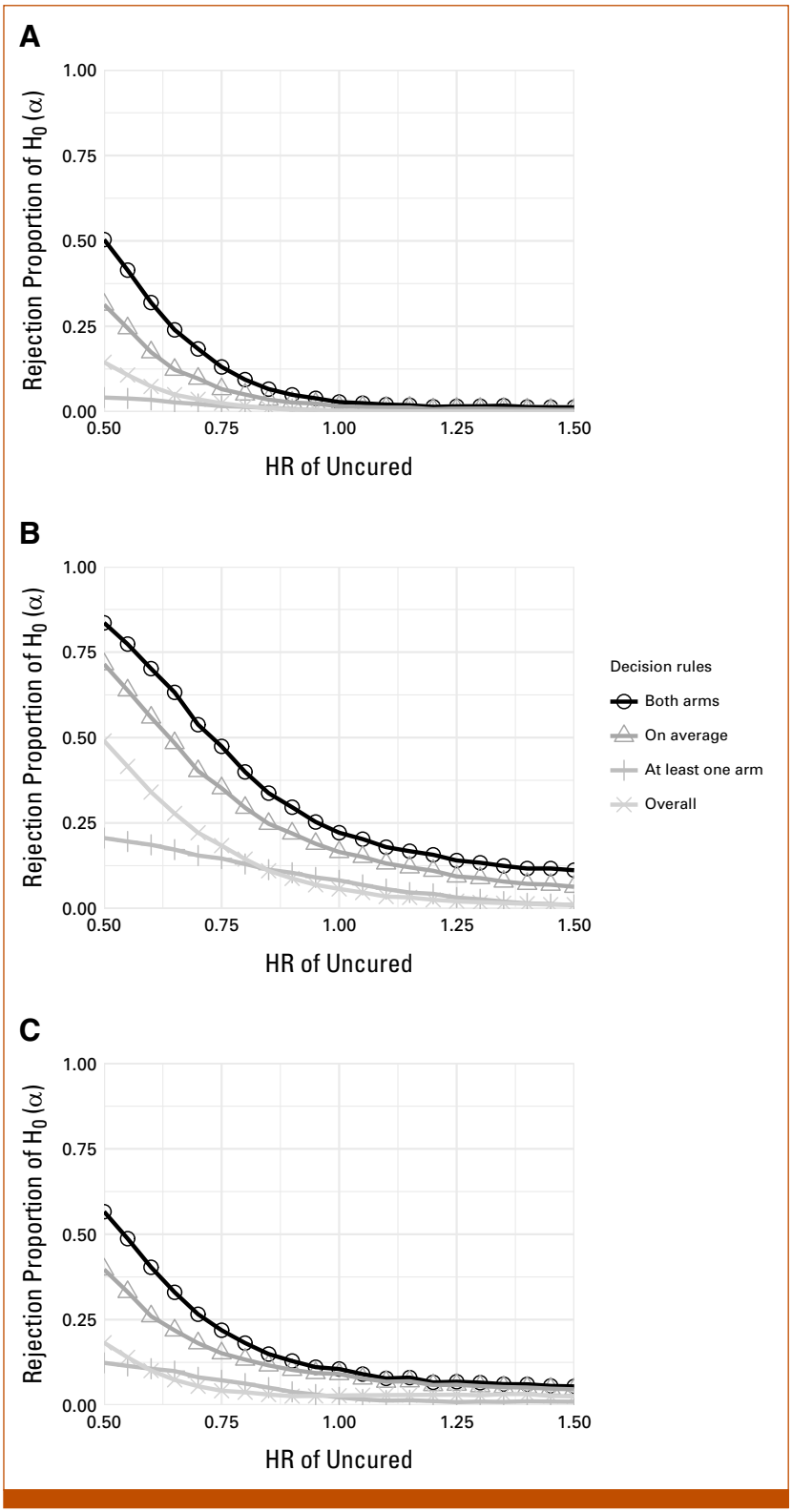

FIG 2. Proportion of rejections of the cure model when the treatment has a small effect on the survival of the uncured, similar to Scenario 3 ($HR_u = 1.06$), with varying effects on cure fraction, generated using a (A) logit, (B) log-log, or (C) probit link function, fitted with a logit link. HR, hazard ratio.

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dropped to 0.5. As above, the use of a log-log function was associated with decreased performance.

Length of Follow-Up

As expected, decreasing follow-up increased the false inference of model inappropriateness, while increasing follow-up improved the performance of the approach (Table 3). Regardless of the setting, performance worsened with the use of a log-log link.

Sample Size

The results were similar across sample sizes (Data Supplement, Tables S3 and S4).

Violation of the Proportional Hazards Assumption

When the hazards were nonproportional, the error rates decreased (Data Supplement, Table S5). This was more pronounced when the shape of the hazard function increased, that is, when it increased slightly over time in the control group and then increased more rapidly in the treated group. As before, the "at least one arm" or "overall" decision rules resulted in the lowest error rates.

Revisiting the Illustrative Examples

Regardless of the decision rule, cure models were found to be appropriate, with the proportion of cured being above 70% in each trial (Data Supplement, Table S6).

DISCUSSION

The use of cure models to analyze RCT data can provide a better understanding of the effect of treatments in favorable situations where some patients may achieve a cure, leading to a situation of heterogeneity in the population. This is particularly true for new therapies such as CAR-T cells for advanced hematologic malignancies[22] or nivolumab and ipilimumab for melanoma.[8] However, the validity of cure models needs to be rigorously tested.[10] Although statistical methods exist for this purpose, they have shown high false decision rates.[8,9,16] More recently, the RECeUS method has been proposed,[14] which considered a homogeneous population. The aim of this work was to study the RECeUS method, when applied in the presence of a potential source of heterogeneity either in the cure rate or in the survival distribution of susceptible (uncured) individuals. As is typical in standard survival analyses, we assumed that there was no informative censoring. For example, this assumes that patients discontinuing treatment after random assignment would not discontinue follow-up, which would lead to (informative) censored observations.

The first question was how the two criteria (cure rate and sufficient follow-up) could be evaluated in the context of an RCT: overall, or stratified for one or both treatment groups, or averaged. Our results suggest that the best decision rule might be to check compliance with the predefined thresholds of the $\hat{\pi}_n$ and $\hat{r}_n$ statistics in at least one group studied ("at least one" rule). In all cases, the choice of link function appears to have a significant impact on the performance of the cure model, with the log-log link function performing worse than logistic and probit links. When the log-log link is correctly specified, the log-log link continues to have poor performance, but this may warrant further research.

The RECeUS method performs well when the treatment has a detrimental effect on the survival of those who are not cured. When the treatment has a beneficial effect on the survival of those who are not cured, the method performs less well, with relatively high error rates, regardless of the link function. This could be explained by the increase in the number of censored patients in the experimental arm when the proportion of cured patients is already high.

However, the "at least one" rule often concludes that a cure model is appropriate at this time across multiple settings, and we suggest future research to assess the estimation and inferential properties of cure models fitted with moderate follow-up and nonzero cure fraction. This would be important for developing the use of cure models for adaptive trials.

We considered the RCT setting of evaluating whether the use of a cure model could be used in post hoc estimation of the treatment effects on the cure rate and short-term survival of the noncure population. This could also be of interest in a prognostic setting to assess which covariate is associated with cure and/or short-term survival, including the prognostic effect of random assignment stratification factors for RCTs. Furthermore, one could estimate the effect of different covariates on either quantity. Multiarm trials could be considered similarly. In a mathematical proof of the statistical properties of cure models, the authors assumed a sufficient follow-up condition for each element in the covariate space.[4] Our current results might suggest it reasonable for at least one group to have evidence of a cure fraction.

Our study has several limitations. Overall survival was the main outcome measure although cancer-specific survival may be of increased interest in older populations; nevertheless, the RECeUS approach could be extended to this framework. We only examined the performance of the RECeUS approach in not concluding that a cure model was appropriate, without assessing whether it led to the failure to identify a cure model that did not exist; however, this was our situation of interest: the assessment of the relevance of a cure model suspected by the observation of a plateau at the end of the survival curves. We primarily considered the joint assessment of both conditions when concluding that the

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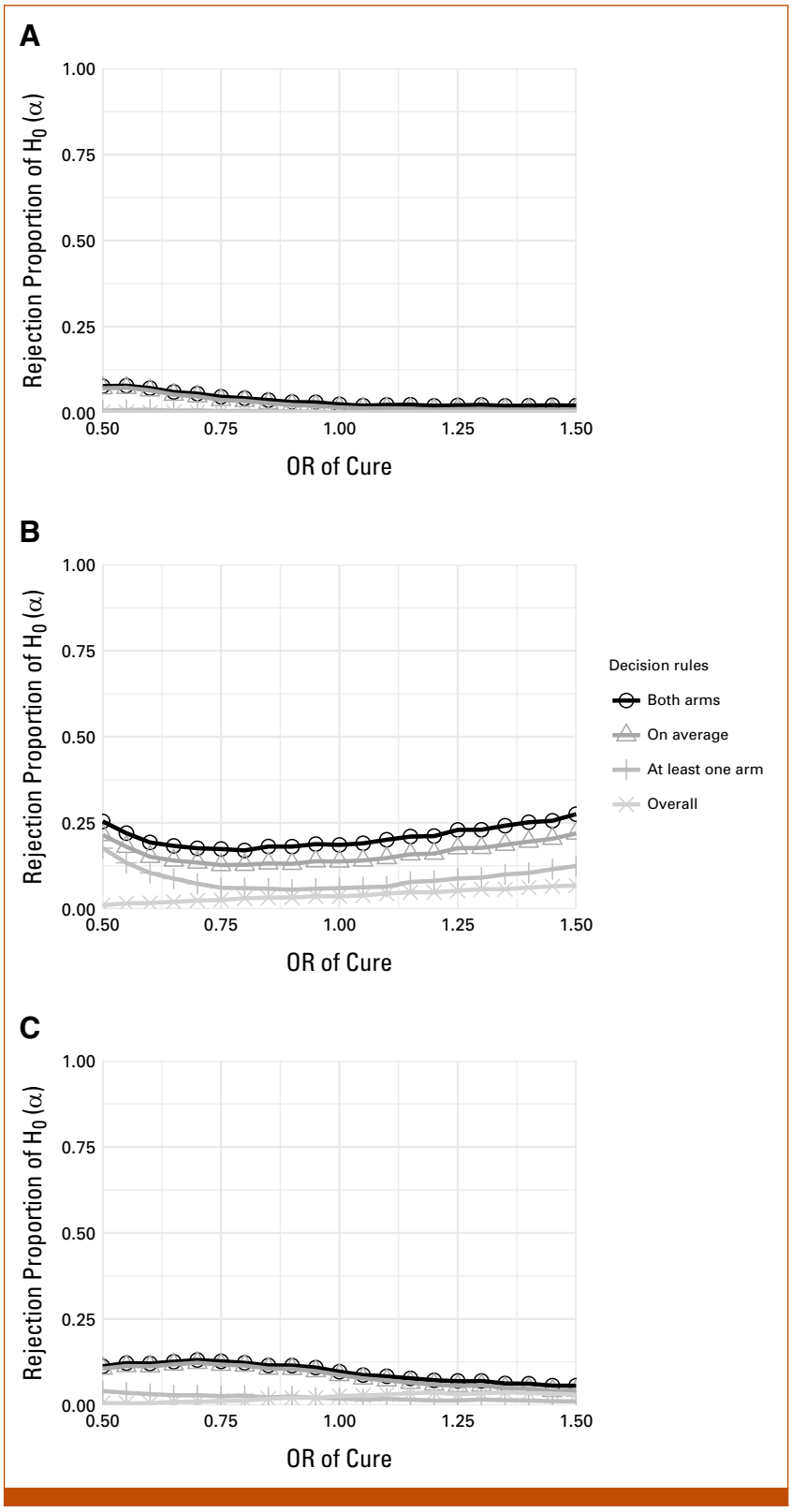

FIG 3. Proportion of rejections of the cure model when the treatment has no effect on the cure fraction, similar to Scenario 3 ($\pi(x = 0) = 0.74$, $OR_{\pi} = 1.06$) but with increasing effect sizes on the survival of the uncured, using a (A) logit, (B) log-log, or (C) probit link function. OR, odds ratio.

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TABLE 3. Simulation Study: Error Rates of Rejecting the Cure Model by Scenario and Decision Rule for Various Lengths of Follow-Up

Decision Rules					
	Both Arms	On Average	At Least One Arm	Overall	
Scenario	Proportion of Cure Model Rejections (95% CI), %				Convergence Rate
24-month data truncation					
Benchmark					
Logit	27.39 (26.52 to 28.27)	18.76 (17.99 to 19.52)	3.85 (3.48 to 4.23)	12.06 (11.42 to 12.70)	99.92
Log-log	60.56 (59.59 to 61.54)	49.83 (48.83 to 50.83)	18.60 (17.82 to 19.37)	40.22 (39.24 to 41.20)	96.79
Probit	38.00 (37.04 to 38.96)	29.99 (29.09 to 30.90)	11.89 (11.25 to 12.53)	24.05 (23.21 to 24.90)	98.82
Scenario 1					
Logit	32.12 (31.21 to 33.04)	24.70 (23.85 to 25.54)	6.01 (5.55 to 6.48)	9.60 (9.02 to 10.17)	99.93
Log-log	61.37 (60.37 to 62.37)	51.70 (50.68 to 52.73)	27.46 (26.55 to 28.38)	25.13 (24.24 to 26.02)	91.33
Probit	32.95 (32.02 to 33.88)	24.92 (24.07 to 25.77)	10.86 (10.25 to 11.47)	11.01 (10.39 to 11.63)	98.72
Scenario 2					
Logit	68.48 (67.57 to 69.39)	55.94 (54.97 to 56.92)	16.21 (15.48 to 16.93)	40.73 (39.77 to 41.69)	99.78
Log-log	91.74 (91.19 to 92.30)	86.34 (85.65 to 87.03)	46.43 (45.43 to 47.44)	75.29 (74.42 to 76.16)	94.44
Probit	74.73 (73.87 to 75.59)	63.95 (63.01 to 64.90)	23.71 (22.87 to 24.56)	48.35 (47.36 to 49.33)	98.21
Scenario 3					
Logit	24.08 (23.24 to 24.92)	15.78 (15.06 to 16.49)	2.66 (2.35 to 2.98)	9.90 (9.32 to 10.49)	99.96
Log-log	58.33 (57.35 to 59.30)	46.94 (45.95 to 47.92)	16.77 (16.03 to 17.51)	36.76 (35.80 to 37.71)	97.88
Probit	33.92 (32.99 to 34.86)	26.14 (25.27 to 27.01)	9.75 (9.16 to 10.33)	22.37 (21.55 to 23.19)	98.78
Scenario 4					
Logit	82.00 (81.24 to 82.75)	69.80 (68.90 to 70.70)	28.92 (28.03 to 29.81)	55.22 (54.24 to 56.20)	99.71
Log-log	93.53 (93.01 to 94.06)	87.34 (86.63 to 88.05)	48.87 (47.80 to 49.93)	80.58 (79.74 to 81.43)	84.25
Probit	86.50 (85.82 to 87.17)	78.09 (77.28 to 78.91)	47.19 (46.21 to 48.18)	66.45 (65.52 to 67.39)	98.64
96-month data truncation					
Benchmark					
Logit	0.01 (0.00 to 0.03)	0.01 (0.00 to 0.03)	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	100
Log-log	3.61 (3.25 to 3.98)	3.60 (3.24 to 3.97)	1.33 (1.10 to 1.55)	0.00 (0.00 to 0.00)	99.38
Probit	1.07 (0.87 to 1.27)	1.04 (0.84 to 1.24)	0.00 (0.00 to 0.00)	0.39 (0.27 to 0.51)	100
Scenario 1					
Logit	0.91 (0.72 to 1.10)	0.91 (0.72 to 1.10)	0.01 (0.00 to 0.03)	0.00 (0.00 to 0.00)	99.99
Log-log	14.12 (13.43 to 14.82)	13.97 (13.27 to 14.66)	11.76 (11.12 to 12.41)	0.00 (0.00 to 0.00)	96.24
Probit	2.05 (1.77 to 2.33)	2.04 (1.76 to 2.32)	0.23 (0.14 to 0.32)	0.90 (0.72 to 1.09)	99.92
Scenario 2					
Logit	0.66 (0.50 to 0.82)	0.18 (0.10 to 0.26)	0.09 (0.03 to 0.15)	0.01 (0.00 to 0.03)	100
Log-log	15.66 (14.94 to 16.38)	10.08 (9.49 to 10.68)	3.31 (2.95 to 3.66)	0.49 (0.35 to 0.63)	98.29
Probit	2.89 (2.56 to 3.22)	2.10 (1.82 to 2.38)	1.60 (1.35 to 1.85)	0.04 (0.00 to 0.08)	99.97
Scenario 3					
Logit	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	0.00 (0.00 to 0.00)	100
Log-log	4.21 (3.82 to 4.61)	4.19 (3.80 to 4.59)	1.10 (0.89 to 1.30)	0.00 (0.00 to 0.00)	99.42
Probit	0.61 (0.46 to 0.76)	0.61 (0.46 to 0.76)	0.02 (-0.01 to 0.05)	0.47 (0.34 to 0.60)	99.99
Scenario 4					
Logit	0.97 (0.78 to 1.16)	0.57 (0.42 to 0.72)	0.18 (0.10 to 0.26)	0.00 (0.00 to 0.00)	100
Log-log	13.42 (12.74 to 14.09)	12.07 (11.42 to 12.71)	2.73 (2.41 to 3.06)	0.23 (0.14 to 0.33)	98.02
Probit	11.82 (11.19 to 12.46)	11.45 (10.83 to 12.08)	5.12 (4.69 to 5.55)	0.66 (0.50 to 0.82)	99.64

cure model was appropriate; however, we also observe that the insufficient follow-up component was the primary criterion leading to the conclusion that a cure model was inappropriate. Finally, we applied the decision thresholds suggested in the original publication as they appear to be reasonable for oncology settings,[14] but the study team could use available information to develop targeted simulations for their study that could update the thresholds.

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In conclusion, the RECeUS approach seems to be of primary interest for detecting a pattern of cure, even in the presence of a possible treatment-related fraction of cure. Its use in randomized trials deserves to be disseminated. Validation of the RECeUS conditions in at least one randomized arm should be sufficient to assess the appropriateness of cure modeling.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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